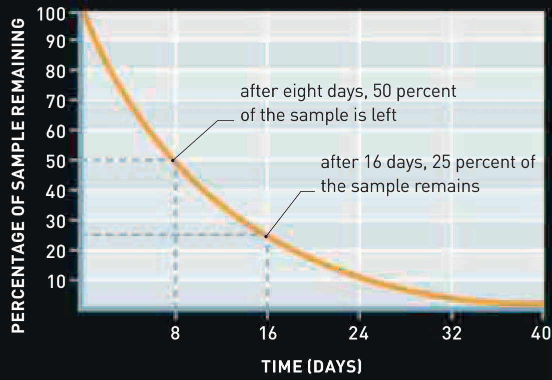
RADIOACTIVE DECAY

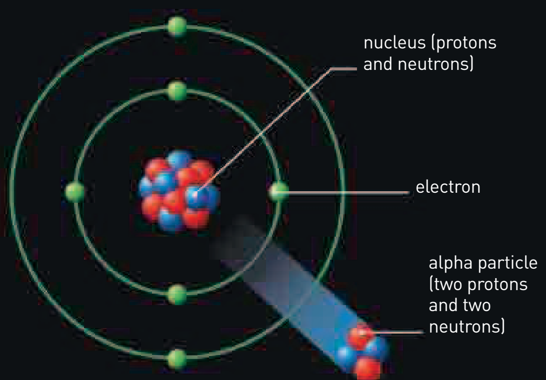
At some point, an unstable nucleus will disintegrate, or "decay." The two most common types of decay are called alpha decay and beta decay (see below). In each case, the nucleus always has excess energy to lose, and that energy is carried away by very short-wavelength, high-energy gamma radiation. The probability that a particular atom will decay is fixed, but there is no way of telling when it will happen. However, in a sample containing a large number of atoms of the same radioactive element, it always takes exactly the same amount of time for half the atoms to decay; that period is known as the element's half-life.



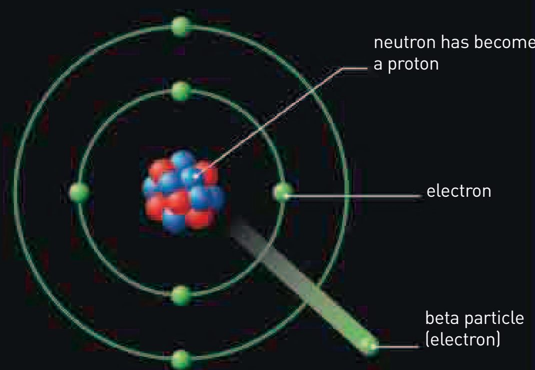
HALF-LIFE
This graph shows the decay curve of a sample of a radionuclide with an eight-day half-life. Every eight days, the number of atoms of that radionuclide halves.

EFFECTS OF RADIOACTIVITY

Radioactivity generates heat, and this is harnessed by the radioisotope thermal generators that power unmanned space satellites and probes. Radioactive substances pose a threat to health because their ionizing effect can damage chemical bonds in compounds essential to life. This can cause a general malaise, called radiation sickness. Damage to DNA inside cells can cause mutations that can result in cancers. Ironically perhaps, radioactive substances are also used in medicine—particularly in radiation therapy to treat cancer.



ALPHA DECAY
An unstable nucleus jettisons a particle made up of two protons and two neutrons: an alpha particle. This makes the nucleus smaller, which sometimes results in greater stability.



BETA DECAY
Inside some unstable nuclei, a neutron may spontaneously become a proton, emitting an electron at the same time. In this case, the electron is known as a beta particle.

RADIATION THERAPY
During radiation therapy, radiation is used to kill cancer cells. The radiation must be carefully controlled because it can damage healthy cells as well as cancerous ones.

RADIOACTIVE DECAY CHAIN
Elements are defined by the number of protons in the nuclei of their atoms. Decay changes the proton number, so the "daughter" nucleus is of a different element. Often the daughter nucleus is also radioactive, and sometimes the process continues in a decay chain like the one shown here.

5.6

OUNCES OF **POTASSIUM** ARE **CONTAINED** IN THE **HUMAN BODY** AND AROUND **4,400** ATOMS **DECAY** PER SECOND